

DSC2204 – Integrative Team Project

Project Title: Timetabling System for SIT Engineering Cluster

1. Problem Definition and Context

This project addresses a real-world academic timetabling problem faced by SIT each semester. The scheduling task requires the simultaneous allocation of multiple interdependent resources, such as rooms, tutors, student groups, and time slots, under numerous operational constraints. Errors or conflicts in scheduling lead to complaints, inefficient resource utilisation, and repeated manual adjustments.

From an operations and supply chain perspective, academic timetabling is a complex resource allocation and constraint satisfaction problem. The difficulty arises from the coexistence of: 1) Hard constraints that must never be violated; 2) Soft constraints representing preferences and quality-of-service considerations, and 3) Data irregularities such as public holidays, odd/even week courses, and mixed delivery modes.

The project focuses initially on the Engineering cluster before extending to the wider all SIT clusters, ensuring both feasibility and scalability.

2. Project Objectives

The overarching objective is to design and demonstrate an intelligent timetabling system that supports decision-making in academic scheduling.

Specific objectives are to:

- 1) Accurately model all relevant scheduling resources and constraints
- 2) Detect and report violations of hard and soft constraints
- 3) Automatically generate feasible timetables that satisfy all hard constraints
- 4) Improve timetable quality through systematic optimisation of soft constraints
- 5) Produce outputs that are interpretable and useful for different stakeholders
- 6) Propose improvements to the existing operational workflow

3. Methodology and System Design

The project methodology is structured into six assessable components:

3.1 Data Modelling

The team will design structured representations for: Courses (delivery mode, enrolment size, odd/even week patterns); Rooms (capacity, physical or virtual); Tutors (availability, workload); Student groups; Time slots and academic calendar elements. The data model is designed for clarity, extensibility, and consistency across programmes.

3.2 Constraint Engine

A constraint engine will be developed to systematically evaluate a timetable against:

1) Hard constraints, including

- No double-booking of rooms, tutors, or student groups
- Room capacity requirements
- Compatibility between delivery mode and room type
- Compliance with academic calendar restrictions

2) Soft constraints, including:

- Minimisation of excessive tutor idle time
- Avoidance of long consecutive student teaching hours
- Reduction of frequent transitions between online and face-to-face modes
- Improvement of campus attendance efficiency

The engine will explicitly distinguish between rule violations and preference trade-offs.

3.3 Schedule Generator

The schedule generator will automatically assign classes to valid time slots and rooms such that all hard constraints are satisfied. The approach will prioritise feasibility and scalability over brute-force enumeration, reflecting operational realism.

3.4 Optimisation Strategy

Given a feasible timetable, the optimiser will iteratively improve solution quality by reducing soft constraint violations using heuristic or rule-based methods. The optimisation rationale will be explained clearly in terms of operations efficiency rather than algorithmic novelty.

3.5 Interface and Outputs

Outputs will be tailored to stakeholders such as programme leaders and administrators. These may include visual timetable views, exportable schedules (e.g. spreadsheet or web-based formats) and summary reports highlighting remaining trade-offs.

3.6 Operational Process Improvement

The team will analyse the current manual scheduling workflow and propose improvements to data consolidation, communication flow, and decision transparency. Assessment emphasis: Logical system decomposition, justification of methods, and operational thinking.

4. Use of AI Tools

Large Language Model (LLM) tools may be used to support ideation, coding, or documentation. However, all outputs generated using AI tools must be critically reviewed and the team must demonstrate understanding of all design decisions. AI usage must be transparent and explicitly documented. The project will be evaluated on reasoning quality, not automation reliance. Assessment emphasis: Accountability, reflection, and integrity in AI-supported work.

5. Deliverables and Assessment Alignment

The project will be assessed based on the following deliverables:

- 1) Functional Prototype: Demonstrates data modelling, constraint checking, schedule generation, and optimisation
- 2) Written Report including problem analysis, methodology and system design, evaluation of outcomes and reflection on limitations and learning
- 3) AI Usage Documentation: Description of how AI tools were used and validated
- 4) Final Presentation: Clear communication of problem, approach, and results

6. Expected Learning Outcomes

Upon completion, students are expected to demonstrate:

- 1) Application of operations and supply chain concepts to real institutional problems
- 2) Ability to model and manage complex constraints
- 3) Team-based problem solving and system integration
- 4) Critical evaluation of solution quality and trade-offs
- 5) Effective communication of analytical results to non-technical stakeholders